SIMATS ENGINEERING

CO2 - ASSESSMENT TOOL 1 - Analytical Problem Solving

COURSE CODE: CSA5103
COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Analytical Problem Solving: 40%

QUESTIONS: **Total Marks: 50**

Annexure A

Code of Conduct:

I RAKSHITHRR (192326156) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Rakshitha BR

Annexure A
Analytical Problem Solving

1. RSA Encryption Analysis
Given p=11, q=13

a. Compute n $\phi(n)$\
b. Choose e=7 and find the private key d
c. Encrypt the message M=9
d. Decrypt the ciphertext and verify the result
e. Analyze how changing e affects security

2. Block Cipher Mode Evaluation
Given plaintext blocks:
P=[1010,1010,1111,1010], Key = K=1100

a. Encrypt using ECB and CBC modes (Assume IV = 0000)
b. Compare the ciphertext outputs
c. Analyze which mode hides patterns better and explain why

3. DES vs 3-DES vs Blowfish Performance Study
Given:

a. DES key size = 56 bits, time = 2 ms/block
b. 3-DES key size = 168 bits, time = 6 ms/block
c. Blowfish key size = 128 bits, time = 1 ms/block

For 1000 blocks of data:
i.Calculate total encryption time for each algorithm ii.Analyze trade-offs between security and performance
iii.Recommend the best algorithm for secure real-time communication

4. Diffie-Hellman Key Exchange Analysis
Given:

a. p=23, g=5
b. User A private key = 6
c. User B private key = 15
d. Compute public keys for A and B
e. Compute the shared secret key
f. Demonstrate how a Man-in-the-Middle attacker could alter the exchange
g. Suggest a secure method to prevent the attack

5. ECC vs RSA Comparison
Given:

a. RSA key size = 1024 bits
b. ECC key size = 160 bits (equivalent security)
c. Device processing capacity = limited (IoT device)
d. Analyze computational cost for both systems
e. Estimate which consumes less power and memory
f. Justify which cryptosystem is better for the given scenario with reasoning

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginnin g)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/form ula with strong justification	Selects correct method with minor errors	Inappropriate e or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganize d steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculation s
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

CSA5103

CRYTOGRAPHY AND NETWORKSECURITY

CO2-AT-1

ANALYTICAL PROBLEM SOLVING

R.R.RAKSHITH
192341156.

5) ECC VS RSA Comparison:-

Gm:-

a) RSA key size = 1024 bits

b) ECC key size = 160 bits (equivalent security)

c) Device processing capacity = limited (IoT device)

d) Analyze computational cost for both systems.

e) Estimate which consumes less power and memory.

f) Justify which cryptosystem is better for the given scenario with reasoning.

Sol:-

Gm:-

- RSA key size = 1024 bits
- ECC key size = 160 bits
- Device = IoT device with limited processing capacity.
- Assumption in the question: both key sizes provide equivalent security.

The objective is to compare the computational cost, power consumption, memory requirements, and suitability of RSA and ECC for a resource-constrained Internet of Things (IoT) device.

a) Analyze the computational cost of RSA and ECC

1) RSA (Rivest-Shamir-Adleman)

RSA is an asymmetric cryptographic algorithm based on modular arithmetic and the mathematical difficulty

of factoring a large composite number

Its public and private operations are based on modular exponentiation.

$$C = M^e \bmod n$$

$$M = C^d \bmod n.$$

Where n is the RSA modulus, e is the public exponent, and d is the private exponent.

Computational cost:-

•) RSA encryption can be relatively fast when a small public exponent is used.

•) RSA decryption generally requires more computation because the private exponent is much larger.

•) Large integer multiplication and modular exponentiation can be demanding for a low-power IoT processor.

For a 1024-bit RSA key, this device must perform operations involving integers up to approximately 1024 bits.

2) ECC (Elliptic Curve Cryptography)

ECC is an asymmetric cryptographic technique based on elliptic curves over finite fields.

Its main operation is

$$Q = kP.$$

where P is a curve point, k is a private scalar, and Q is the resulting public point.

ECC uses smaller keys and is generally suitable for resource-constrained IoT devices.

(a) Comparison of Computational cost:-

Parameter	RSA (1024-bit)	ECC (160-bit)
Main Operation	Modular exponentiation	Scalar multiplication
key Size	1024 bits	160 bits
Processing cost	Generally higher for private operations	often lower overall.
Resource usage	More demanding	More efficient.

Interpretation:- ECC generally requires fewer computational and storage resources, although actual performance depends on the implementation and hardware

(b) Power and memory consumption:-

Memory Calculation:-

RSA key size :-

$$1024/8 = 128 \text{ bytes}$$

ECC keysize:-

$$160/8 = 20 \text{ bytes}$$

Percentage reduction :-

$$\frac{128-20}{128} \times 100 = 84.375\%$$

ECC requires less key storage. It can also consume less energy because its operations are often more efficient on constrained devices. Actual

power consumption depends on the hardware and implementation.

(c) Best cryptosystem for IoT devices:-

ECC is generally more suitable for IoT devices because it offers:-

•) Lower memory usage: Smaller keys require less storage.

•) Lower Communication overhead:- Public keys are smaller.

•) Efficient processing :- Often requires fewer resources.

•) potential power savings:.. Can reduce energy consumption in battery-powered devices.

For example, a battery-powered temperature sensor can use ECC for secure key exchange with a cloud server, while AES encrypts the actual sensor data.

Conclusion:- ECC is generally preferable for resource-constrained IoT devices because of its smaller keys and efficient resource usage. However, the given 1024-bit RSA and 160-bit ECC keys sizes are outdated for modern security; real systems should use currently recommended parameters and secure protocols.

1) question 1: RSA Encryption Analysis:

Given:

$p = 11$

$q = 13$

Public exponent $e = 7$

plaintext message $M = 9$.

(a) Compute n and $\phi(n)$:-

Step-1:- Calculate the modulus n

In RSA, the modulus is the product of two prime numbers.

formula:-

$$n = p \times q$$

substituting the given values :-

$$n = 11 \times 13$$

$$\boxed{n = 143.}$$ //

Step-2 :- Calculate Eulers totient function $\phi(n)$

Since p and q are distinct prime numbers:

$$\phi(n) = (p-1)(q-1)$$

Substituting

$$\phi(n) = (11-1)(13-1)$$

$$\phi(n) = 10 \times 12$$

$$\boxed{\phi(n) = 120.}$$ //

Result:-

*) RSA modulus $n=143$

*) Euler's totient $\phi(n)=120$

The modulus n is used in both encryption and decryption. The totient $\phi(n)$ is used to calculate the private exponent.

(b) Choose $e=7$ and find the private key d

The private exponent d must satisfy the modular inverse relationship :-

$$d \times e \equiv 1 \pmod{\phi(n)}$$

Substituting $e=7$ and $\phi(n)=120$:-

$$7d \equiv 1 \pmod{120}$$

This means we must find an integer d such that $7d$ leaves a remainder of 1 when divided by 120.

Step-1:- Verify that e is valid.

For RSA, e must satisfy :-

$$\gcd(e, \phi(n)) = 1.$$

Here :

$$\gcd(7, 120) = 1.$$

Step-2:- Find the modular inverse using the Extended Euclidean Algorithm :-

$$-48^{0} \times 48^{32} \times 48^{4} \times 48^{2} \times 48.$$

First, divide -120 by 7:

$$120 = 7 \times 17 + 1$$

Rearrange the equation

$$1 = 120 - 7 \times 17$$

$$\therefore \quad 1 = 120 + 7(-17)$$

This gives

$$7(-17) \equiv 1 \pmod{120}$$

Since $-17 \equiv 103 \pmod{120}$, the positive modular inverse is:-

$$\boxed{d = 103}$$

verification:-

$$e \times d = 7 \times 103 = 721$$

$$721 = 6 \times 120 + 1$$

Hence

$$721 \bmod 120 = 1$$

The modular inverse is correct.

Final RSA keys:-

-) Public key : $(e, n) = (7, 143)$
-) Private key : $(d, n) = (103, 143)$

The public key is used for encryption, while the private key is used for decryption.

(e) Encrypt the message $M = 9$:-

RSA encryption uses the formula:-

$$C = M^e \bmod n.$$

Given:

$$M = 9, \quad e = 7, \quad n = 143.$$

Substituting

$$C = 9^7 \bmod 143.$$

Step 1: Calculate powers of 9 modulo 143

$$9^2 = 81$$

$$9^3 = 9 \times 81 = 729$$

Since $729 = 5 \times 143 + 14$:.

$$9^3 \bmod 143 = 14$$

Next:-

$$9^4 \bmod 143 = (9^3 \bmod 143) \times 9 \bmod 143$$
$$= 14 \times 9 \bmod 143 = 126$$
$$9^5 \bmod 143 = 126 \times 9 \bmod 143$$
$$= 1134 \bmod 143 = 132$$
$$9^6 \bmod 143 = 132 \times 9 \bmod 143$$
$$= 1188 \bmod 143 = 43.$$

Finally:

$$9^7 \bmod 143 = 43 \times 9 \bmod 143.$$
$$= 387 \bmod 143.$$

Since $387 = 2 \times 143 + 101$:

$$C = 101.$$

Final answer: The ciphertext is 101.

(d) Decrypt the ciphertext and verify the Result:

RSA decryption uses the formula:

$$M = C^d \bmod n$$

From part (c), the correct ciphertext is $C = 48$. The private exponent is $d = 103$, and $n = 143$.

Thus: $M = 48^{103} \bmod 143$

To simplify calculation, use repeated squaring

step-1:- Calculate successive powers modulo 143.

$48^2 = 2304 \bmod 143 = 16$

$48^4 = 16^2 = 256 \bmod 143 = 113$

$48^8 = 113^2 \bmod 143 = 42$

$48^{16} = 42^2 \bmod 143 = 48$

$48^{32} = 48^2 \bmod 143 = 16.$

$48^{64} = 16^2 \bmod 143 = 113.$

step-2:- Express the exponent as a sum of powers 2

$$103 = 64 + 32 + 4 + 2 + 1.$$

$$\therefore 48^{103} = 48^{64} \times 48^{32} \times 48^4 \times 48^2 \times 48.$$

Taking each power modulo 143 :-

$$M = (113 \times 16 \times 113 \times 16 \times 48) \bmod 143.$$

Simplifying step by step :-

$113 \times 16 = 1808 \bmod 143 = 92$

$92 \times 113 = 10396 \bmod 143 = 100$

$100 \times 16 = 1600 \bmod 143 = 27$

$27 \times 48 = 1296 \bmod 143 = 9.$

Hence $\boxed{M = 9.}$

Verification

-) Original Message : $M = 9$

-) Encrypted ciphertext $C = 48$

-) Decrypted plaintext : $M = 9.$

Since the decrypted message matches the original message, RSA encryption and decryption are verified.

Arithmetic correction part c) $95 \bmod 143 = 133,096$ mod 143 = 53 and $97 \bmod 143 = 48$. The Correct ciphertext is 48, as used in this decryption

(e)) Analyze how changing e affects security.

The public exponent e determines how the plaintext is transformed during RSA encryption:

$$C = M^e \bmod n.$$

However, e cannot be chosen arbitarily, It must satisfy:

$$1 < e < \phi(n)$$

and

$$\gcd(e, \phi(n)) = 1$$

For this example, $\phi(n) = 120$.

Public exponent e	Validity	Analysis
$e = 3$	Valid	fast encryption
$e = 7$	Valid	Own exponent $d = 103$
$e = 10$	Invalid	$\gcd(10, 120) \neq 1$
$e = 17$	Valid	Has a modular inverse modulo 120
$e = 120$	Valid	Violates $e < \phi(n)$. and is not coprime to 120.

Conclusion:-

Choosing e affects encryption efficiency. and the mathematical relationship b/w the public and private keys. A valid exponent is necessary, but security cannot be judged by the size of e alone.